

Descriptive Statistics (Part 2)

Session: Understanding Data Variability (Spread of Data)

1. Why Do We Need Measures of Spread?

Two datasets can have the same **mean** but look completely different.

- **Dataset A:** 11, 12, 12, 13 (Values are bunched together)
- **Dataset B:** 2, 5, 12, 29 (Values are spread far apart)

Measures of Dispersion describe how far values are spread from the center.

2. Range, Variance, and Standard Deviation

Range

The simplest measure of spread. It is the difference between the highest and lowest values.

Formula: $\text{Range} = \text{Max} - \text{Min}$

Variance (σ^2)

Measures the average squared distance of each data point from the mean.

- **Small Variance:** Data is close to the mean.

- **Large Variance:** Data is widely spread.

Standard Deviation (σ)

The square root of the variance. It is the most common measure of spread because it is expressed in the **same units** as the original data.

Formula: $\sigma = \sqrt{\sigma^2}$

3. Quartiles and Percentiles

Quartiles

Quartiles divide a sorted dataset into four equal parts:

- **Q1 (25th Percentile):** 25% of data falls below this value.
- **Q2 (50th Percentile):** The **Median**.
- **Q3 (75th Percentile):** 75% of data falls below this value.

Interquartile Range (IQR)

The spread of the **middle 50%** of the data. It is highly effective for identifying outliers.

Formula: $IQR = Q3 - Q1$

4. Detecting Outliers

Outliers are extreme values that can distort your analysis. We use the **IQR Rule** to find them:

- **Lower Bound:** $Q1 - (1.5 \times IQR)$
- **Upper Bound:** $Q3 + (1.5 \times IQR)$
- *Any value outside these bounds is considered an outlier.*

Python

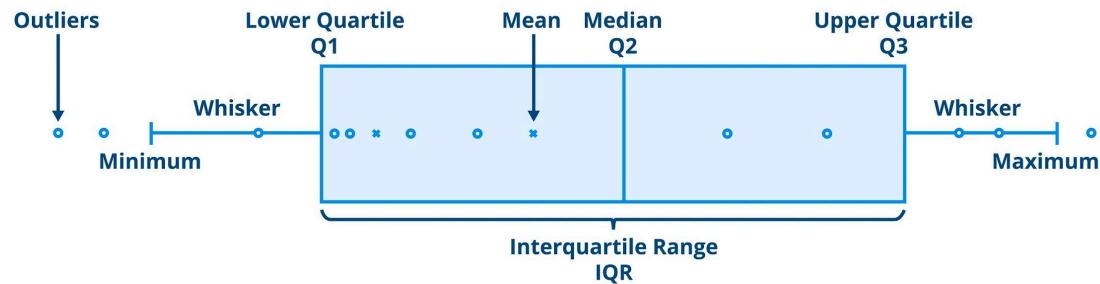
```
1. import numpy as np
2.
3. data = [10, 20, 30, 40, 50, 60, 70, 80, 200]
4. q1, q3 = np.percentile(data, [25, 75])
5. iqr = q3 - q1
6.
7. lower_bound = q1 - 1.5 * iqr
8. upper_bound = q3 + 1.5 * iqr
9.
10. outliers = [x for x in data if x < lower_bound or x > upper_bound]
11. print("Detected Outliers:", outliers)
```

5. Visualization: Boxplots and Histograms

To understand spread visually, we use two primary tools:

1. **Histogram:** Shows the frequency and "shape" of the data distribution.
2. **Boxplot:** Visually highlights the Median, Quartiles, and Outliers.

Box plot



6. Dispersion in Machine Learning

Spread is critical for **Feature Scaling**. Many algorithms (like KNN, SVM, and Neural Networks) perform poorly if features have different scales.

Standardization (Z-score):

This formula converts data so it has a Mean of 0 and a Standard Deviation of 1.

$$z = \frac{x - \mu}{\sigma}$$

Where x = value, μ = mean, σ = standard deviation.

Key Takeaways

1. **Range** gives a quick look at the total spread.
 2. **Standard Deviation** shows the "average" spread around the mean.
 3. **IQR** is the best tool for identifying **outliers**.
 4. **Boxplots** are the standard visual for understanding data variability.
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Homework for Students

Dataset: `data = [12, 15, 18, 20, 22, 25, 30, 35, 40, 100]`

Tasks:

1. Calculate the **Range**.
2. Calculate the **Standard Deviation** using `np.std()`.
3. Find **Q1**, **Median**, and **Q3**.
4. Calculate the **IQR**.
5. Use the **IQR Rule** to determine if `100` is statistically an outlier.
- 12.